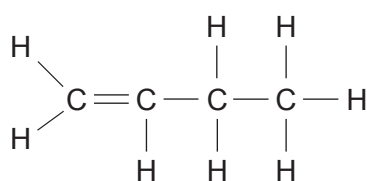
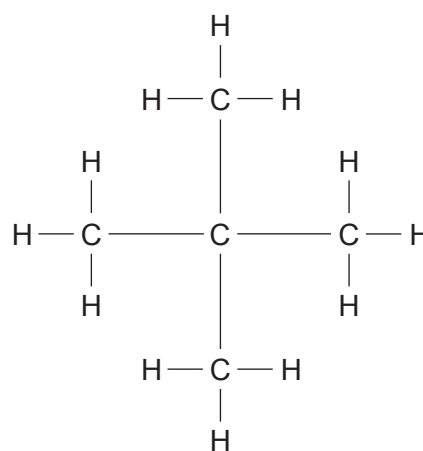
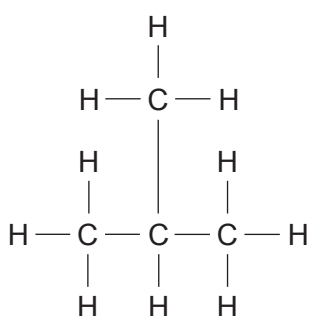
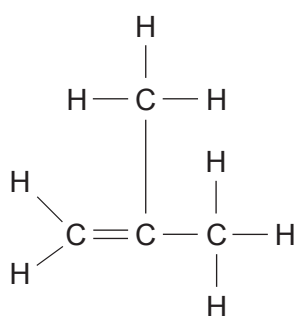
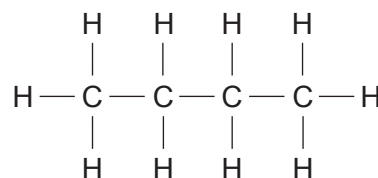
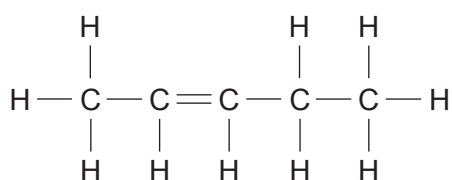
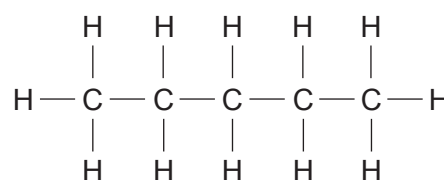


3. (a) The structural formulae of seven organic compounds, **A-G**, are shown below.

**A****B****C****D****E****F****G**

- (i) Give the letters of the **three** compounds that have the general formula C_nH_{2n} . [1]

.....

- (ii) State the **number** of compounds that are saturated. [1]

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- (iii) Give the **letter** of the compound that is but-1-ene. [1]

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- (iv) Draw the structural formula of the isomer of C_5H_{12} **not** shown. [1]

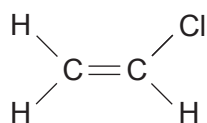
- (v) Describe the chemical test that can be used to distinguish between compounds **C** and **D**.

Give the expected observation for both compounds. [2]

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.....



- (b) The structural formula of vinyl chloride is shown below.



Vinyl chloride can undergo a reaction known as polymerisation to produce polyvinyl chloride, commonly known as PVC.

Explain what happens to vinyl chloride molecules during the formation of PVC. Draw the repeating unit of PVC. [3]

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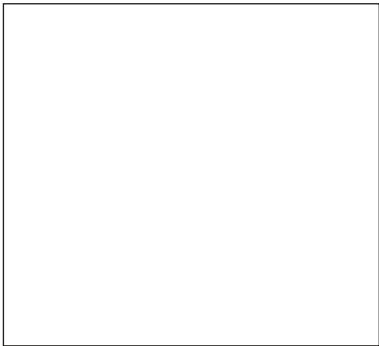
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repeating unit

- (c) PVC is widely used in everyday life and has replaced many traditional materials.

Apart from cost, suggest **two** reasons why PVC has replaced iron for making drainpipes and guttering. [1]

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